Yuyang Zhang

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Education

University of Southern California — **Los Angeles, CA**
M.S., Computer Science, GPA: 4.0/4.0 — *2025/9– 2027/5*

Huazhong University of Science and Technology — **Wuhan, China**
B.Eng., Information Security, GPA:3.81/4.0 — *2021/9– 2025/6*

Experience

Dasoundgen Technology Co, Ltd. — **Wuhan, China**
Software Engineer Intern — *2024/5– 2024/10*

- Designed and implemented a full-stack communication pipeline where audio files from a React frontend are processed by a finetuned neural network backend and returned as structured classification results.
- Developed RESTful APIs using Express, handling file uploads, request validation, and result delivery. Reduced the P95 E2E latency by 30% using binary upload, Multer and model warm-up.
- Deployed the containerized backend (Docker) for the keyword detection system on AWS ECS Fargate behind an ALB, serving an average of 1k+ daily requests, and managed over 1TB of audio assets in S3 with lifecycle policies for cost optimization.

Zhongguancun Laboratory — **Beijing, China(Remote)**
Software Engineer Intern — *2023/3– 2023/11*

- Participated in the development of an integrated fuzzing platform. By decoupling the underlying engine and scheduling logic, achieved modular integration of mainstream testing technologies such as AFL and LibFuzzer.
- Designed and implemented a dynamic policy configuration module, allowing users to flexibly customize mutation algorithms and seed selection strategies based on target program characteristics, significantly enhancing testing flexibility for complex programs.
- Optimized the mutation logic and feedback mechanism of test cases, increasing the system's code coverage on specific benchmark datasets from 68% to 83%, and successfully reproduced multiple potential security vulnerabilities.

Projects

Backdoor Attack Framework on Diffusion based Image Compressors — 2024/10-2025/4

- Designed and implemented an on-the-fly data generation pipeline using Python and multithreading, which dynamically synthesized and augmented training samples in memory, reducing storage requirements by 100% and accelerated model iteration time by 30%.
- Enhanced the robustness of a Diffusion-based image compression model against backdoor attacks by fine-tuning it on both original and poisoned datasets. Subsequently proposed defense mechanisms to prevent malicious image degradation and fortify model security.
- Designed and developed a lightweight web-based GUI with React where users can upload images, trigger compression on both clean and compromised models, and interactively compare results with highlighted degradations, showcasing security risks in a user-friendly manner.

Medical Image Segmentation App — 2023/6-2023/8

- Engineered a web-based medical imaging platform that allows doctors to upload lung CT scans, processes them with a 3D ResUNet model backend, and returns real-time segmentation results.
- Implemented an interactive 3D visualization module on the web frontend with React Three Fiber and OrbitControls, enabling physicians to rotate and inspect segmented CT volumes for more efficient diagnosis.
- Awarded National 3rd Prize at the 2023 National College Student Information Security Contest for developing a system that achieved a mean Dice Coefficient of 0.91 in image segmentation through an innovative integration of AI, privacy-preserving computation, and interactive web technologies.

AI Research Agent — 2026/3-2026/4

- Built an AI-powered research agent leveraging LLMs (OpenAI API) with tool-calling and RAG pipeline, enabling automated information retrieval and multi-step reasoning across web data
- Designed and implemented a scalable backend using FastAPI, integrating vector search (FAISS) and embedding-based retrieval to improve answer relevance and context grounding
- Developed a multi-agent workflow with query planning and parallel tool execution, improving information coverage and response quality for complex user queries

Skills

Python/Pytorch, C/C++, Java, HTML/CSS/Javascript, React, FastAPI,Node.js, MongoDB, SQL, Git, AWS, Docker, Cursor, Claude Code, Codex